1. Assume $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{p-1} x_{ip-1} + \varepsilon_i$, $\varepsilon_i \sim i.i.d. N(0, \sigma^2)$. Let $y = (y_1, \dots, y_n)^T, \varepsilon = (\varepsilon_1, \dots, \varepsilon_n)^T, \beta = (\beta_0, \dots, \beta_{p-1})^T, X = (1, x_1, \dots, x_{p-1})$, where $x_j = (x_{1j}, \dots, x_{nj}), j = 1, \dots, p-1$.

(a) Let the best estimate of β as the one which minimizes

$$\sum_{i=1}^{n} \varepsilon_i^2.$$

Derive the least square estimate (OLS) $\hat{\beta}$.

$$\sum_{i}^{n} \varepsilon_i = \sum_{i}^{n} (y_i - x_i^T \beta) = (Y - X\beta)^T (Y - X\beta) = y^T y - 2y^T X\beta + \beta^T X^T X\beta$$

$$\frac{\partial [y^T y - 2y^T X\beta + \beta^T X^T X\beta]}{\partial \beta} = -2X^T y + 2X^T X\beta$$

$$-2X^T y + 2X^T X\hat{\beta} = 0, \hat{\beta} = (X^T X)^{-1} X^T y$$

(b) Find the distribution of $\hat{\beta}$.

$$y \sim N(X\beta, \sigma^2 I)$$

$\hat{\beta}$ is a linear combination of y, $\hat{\beta}$ follow a multivariate normal distribution

$$E(\hat{\beta}) = E[(X^T X)^{-1} X^T y] = (X^T X)^{-1} X^T E(y) = \beta$$

$$Var((X^T X)^{-1} X^T y) = (X^T X)^{-1} X^T \cdot Var(y) X (X^T X)^{-1} = \sigma^2 (X^T X)^{-1}$$

Therefore, $\hat{\beta} \sim N(\beta,\ \sigma^2 (X^T X)^{-1})$

(c) Write down the equation of R^2 and the unbiased estimate of σ^2.

Let $\hat{y}_i = x_i^T \hat{\beta}$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)}{\sum_{i=1}^{n} (y_i - \bar{y})}$$

$$\hat{\sigma}^2 = \frac{RSS}{n-p} = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)}{n-p}$$

2. Assume $y_i \sim Bernoulli(p_i)$ for $\mathrm{i} = 1, \ldots, \mathrm{n}$, where $p_i = \Pr(y_i = 1|x_i)$.

The logistic regression model is defined as

$$\ln\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_i$$

(a) Show that the probability p_i can be written as the logistic function:

$$p_i = \frac{1}{1+e^{-(\beta_0+\beta_1 x_i)}}$$

$$\ln\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_i$$

$$\frac{p_i}{1-p_i} = e^{\beta_0+\beta_1 x_i}$$

$$p_i = \frac{1}{1+e^{-(\beta_0+\beta_1 x_i)}}$$

(b) Instead of minimizing the sum of squared errors, logistic regression uses Maximum Likelihood Estimation (MLE). Write down the Log-likelihood function $l(\beta_0, \beta_1)$ for this model.

The Likelihood function:

$$L(\beta_0, \beta_1) = \prod_{i=1}^{n}\left[p_i^{y_i}(1-p_i)^{1-y_i}\right] = \prod_{i=1}^{n}\left[\left(\frac{p_i}{1-p_i}\right)^{y_i}(1-p_i)\right]$$

The log-likelihood:

$$l(\beta_0, \beta_1) = \sum_{i=1}^{n}\left[y_i(\beta_0 + \beta_1 x_i) - \log\left(1 + e^{\beta_0+\beta_1 x_i}\right)\right]$$

(c) Define the deviance for this model, and state its asymptotic distribution.

Deviance $= -2l(\beta_0, \beta_1) = -2\sum_{i=1}^{n}\left[y_i(\beta_0 + \beta_1 x_i) - \log\left(1 + e^{\beta_0+\beta_1 x_i}\right)\right]$

3. Consider the dataset fat:

name	Description
fat	body fat
age	Age (years)
weight	Weight (lbs)
height	Height (inches)
knee	Knee circumference (cm)

We fit the following linear regression model (Model A):

$$\text{fat} = \beta_0 + \beta_1 age + \beta_2 weight + \beta_3 height + \beta_4 knee + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2)$$

The output from a multiple linear regression analysis of Model A is provided:

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 22.50145   10.35663   2.173   0.0308 *
age          0.17104    0.02965   5.768 2.39e-08 ***
weight       0.21156    0.02415   8.759 3.21e-16 ***
height      -0.59029    0.10720  -5.507 9.16e-08 ***
knee        -0.19365    0.29268  -0.662   0.5088
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.815
Multiple R-squared: ??
F-statistic:  68.2 on 4 and 247 DF,  p-value: < 2.2e-16
```

(a) Estimate $\hat{\sigma}^2$.

$$(5.815)^2 \approx 33.814$$

(b) How many observations were use in this analysis (i.e., observations with all variables measured)?

$$252 (= 247 + 4 + 1)$$

(c) Interpret the estimated coefficient of age, which is 0.17104.

Holding weight, height, and knee measurements constant, a one-year increase in age is associated with a 0.17104-unit increase in body fat on average.

(d) Write down the null and alternative hypothesis corresponding to the F-statistic. Based on the output, explain the conclusion of this test.

$$H_0: \beta_{age} = \beta_{weight} = \beta_{height} = \beta_{knee} = 0, \qquad H_1: o.w.$$

The p-value $\ll 0.05$, reject H_0. Therefore, the linear model has explanatory power.

Consider an alternative model (Model B) obtained by removing the least significant variable, knee. The regression output is provided below:

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 17.76738    7.47935   2.376   0.0183 *
age          0.16979    0.02956   5.744 2.70e-08 ***
weight       0.19815    0.01313  15.095  < 2e-16 ***
height      -0.59433    0.10690  -5.560 6.97e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.809 on 248 degrees of freedom
Multiple R-squared:  0.524,
```

(e) Based on the analysis results available, can you give the R-squared value of Model A (marked as ??) a reasonable non-zero lower bound without doing any calculations? If yes, explain why it is a lower bound. If no, explain why.

Since Model B is a sub model of Model A, the R^2 for Model B (0.524) is smaller than the R^2 of Model A. Therefore, 0.524 can be a considered a reasonable lower bound for the R^2 of Model A.

4. (30) Consider the dataset **birthwt**. The goal is to study factors associated with low birth weight.

Name	Description
low (Response)	1 if birth weight < 2.5 kg, 0 otherwise
age	Mother's age (years)
lwt	Mother's weight (lbs)
smoke	1 if mother smokes, 0 otherwise
ht	1 if mother has hypertension, 0 otherwise
ui	1 if uterine irritability present, 0 otherwise

Output from the logistic regression model:

```
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.399794   1.080407   1.296   0.1951
age         -0.034073   0.033674  -1.012   0.3116
lwt         -0.015447   0.006587  -2.345   0.0190 *
smoke        0.647540   0.336650   1.923   0.0544 .
ht           1.893274   0.683392     ??    0.0056 **
ui           0.884607   0.444051   1.992   0.0464 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 234.67  on 188  degrees of freedom
Residual deviance: 211.78  on 183  degrees of freedom
AIC: 223.78
```

(a) (3) Under the null hypothesis $H_0: \beta_{ht} = 0$. Write down the test statistic (marked as ??) and its distribution.

The test statistic is $\frac{1.893274}{0.444051} \approx 4.264$. This test statistic follows a standard normal distribution, $N(0,1)$.

(b) (4) Calculate the Odds Ratio (OR) for low birth weight comparing mothers who both smoke and have hypertension with those who neither smoke nor have hypertension, while keeping all other variables unchanged.

$$OR = \exp(0.647540 + 1.893274) = \exp(2.540814) \approx 12.69$$

(c) (4) Interpret the estimated coefficient of age, which is -0.034. Is this predictor statistically significant at the $\alpha = 0.05$ level?

Holding the other variables (lwt, smoke, ht, ui) constant, a one-year increase in the mother's age is associated with a decrease of 0.034 in the log-odds of low birth weight. P-value = 0.31 > 0.05, ∴ not statistically significant.

(d) (3) Write down the estimated linear predictor equation for the log-odds: $\hat{\eta} = \log\left(\frac{\hat{p}}{1-\hat{p}}\right)$.

$$\hat{\eta} = \log\left(\frac{\hat{p}}{1-\hat{p}}\right)$$
$$= 1.399794 - 0.034073(age) - 0.015447(lwt)$$
$$+ 0.647540(smoke) + 1.893274(ht) + 0.884607(ui)$$

(e) (3) Calculate the Likelihood Ratio Test (LRT) statistic to test the global null hypothesis ($H_0 = \beta_{Adhes} = \beta_{BNucl} = \cdots = \beta_{USize} = 0$). What is the distribution of the test statistic under H_0?

LRT statistic $= 234.67 - 211.78 = 22.89$, the null distribution is $\chi^2_{df=5}$

(f) (3) Suppose we perform backward variable selection using the R command drop1(). Based on the output, which variable is the best candidate for removal in the first step? Justify your choice?

```
Model:
low ~ age + lwt + smoke + ht + ui
       Df Deviance    AIC
<none>      211.78 223.78
age     1   212.83 222.83
lwt     1   218.04 228.04
smoke   1   215.48 225.48
ht      1   219.84 229.84
ui      1   215.68 225.68
```

The best candidate for removal is age, since dropping age results in a smaller AIC (222.83 < 223.78), and it yields the lowest AIC among all candidate models.

(g) (5) Suppose we evaluate the current model (prior to variable selection) using an independent dataset. The observed outcomes (y) and their corresponding predicted probabilities ($\hat{p}$) are shown in the table below:

low (y)	**1**	**0**	**1**	**0**	**0**	**1**	**0**
$\hat{p}$	0.82	0.74	0.66	0.41	0.29	0.71	0.33

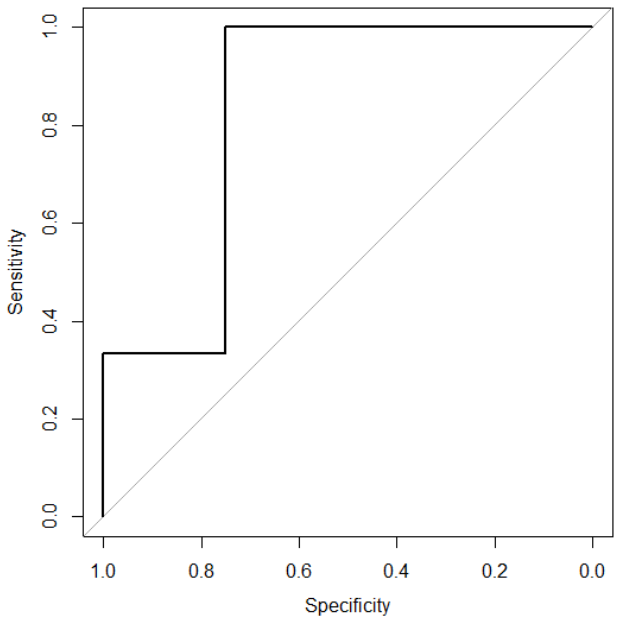
Using a classification cutoff of $c = 0.5$ (where $\hat{y} = 1$ if $\hat{p} \geq 0.5$. $p = 0.5$, calculate the sensitivity and specificity of the model.

	$y = 1$	$y = 0$
$\hat{p} > 0.5$	3	1
$\hat{p} \leq 0.5$	0	3
Total	3	4

Sensitivity = 1

Specificity = 3/4

(h) (5) Based on the data provided in part (g), plot the ROC curve, and compute the Area Under the Curve (AUC).



AUC $= \frac{1}{4} \cdot \frac{1}{3} + \frac{3}{4} \cdot 1 = \frac{5}{6} \approx 0.83$

5. (20) The sinking of the Titanic is a well-known historical event. A study was conducted to investigate the association between **sex** and **survival** among passengers. A total of 400 individuals were selected for each survival group:

The distribution of passengers by sex and survival status is given below:

	alive	dead	Total
Female	300	100	400
Male	100	300	400
Total	400	400	800

(a) (3) Is this a prospective or retrospective study? Briefly explain your answer.

This is a retrospective study. The numbers of cases (alive) and controls (dead) are determined in advance, and the sampling rates for cases and controls are different.

(b) (5) Our goal is to estimate the odds ratio (OR) to understand how sex is associated with survival:

$$\mathrm{OR} = \frac{\dfrac{\Pr(alive|Female)}{\Pr(dead|Female)}}{\dfrac{\Pr(alive|Male)}{\Pr(dead|Male)}}$$

To evaluate the OR in the above formula, do we need to estimate $\Pr(alive|Female)$ and $\Pr(alive|Male)$ first? Can we estimate these probabilities by this dataset? If not, explain how we can still compute $\widehat{\mathrm{OR}}$.

We cannot directly estimate $\Pr(alive|Female)$ or $\Pr(alive|Male)$ from this dataset because the sampling is retrospective. However, we can estimate $\Pr(Female|alive)$ and $\Pr(Male|alive)$. The OR can be estimated by the following equation:

$$OR = \frac{\dfrac{\Pr(alive|Female)}{1-\Pr(alive|Female)}}{\dfrac{\Pr(alive|Male)}{1-\Pr(alive|Male)}} = \frac{\dfrac{\Pr(Female|alive)}{1-\Pr(Female|alive)}}{\dfrac{\Pr(Female|dead)}{1-\Pr(Female|dead)}}$$

Therefore, $\widehat{OR} = \dfrac{\frac{\frac{300}{400}}{1-\frac{300}{400}}}{\frac{\frac{100}{400}}{1-\frac{100}{400}}} = 9$

(c) (9) Suppose that a logistic regression model is fitted to the data, with alive as the binary response variable and sex the explanatory variable. Let $p_i = \Pr(alive|sex_i)$.

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 sex_i$$

Treatment coding is used for the variable sex (i.e., 0=Female, 1=Male). Based on the log-likelihood given in Question 2, derive the score function and find the MLE of the parameter β_1 (the estimated value of the coefficient of sex). Is this MLE estimate consistent with the odds ratio obtained in part (b)?

$$l(\beta_0, \beta_1) = \sum_{i=1}^{n} \left[y_i(\beta_0 + \beta_1 x_i) - \log\left(1 + e^{\beta_0 + \beta_1 x_i}\right) \right]$$

The score function:

$$\frac{\partial l(\beta_0, \beta_1)}{\partial \beta_0} = \sum_{i=1}^{n} y_i - \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$$

$$\frac{\partial l(\beta_0, \beta_1)}{\partial \beta_1} = \sum_{i=1}^{n} y_i x_i - \frac{x_i e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$$

(y_i, x_i)	Count
$(0,0) = (\text{dead}, \text{female})$	100
$(0,1) = (\text{dead}, \text{male})$	300
$(1,0) = (\text{alive}, \text{female})$	300
$(1,1) = (\text{alive}, \text{male})$	100

$$\frac{\partial l(\beta_0, \beta_1)}{\partial \beta_1} = \sum_{i=1}^{n} y_i x_i - \frac{x_i e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$$

$$= 100(0-0) + 300\left(0 - \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}}\right) + 300(0-0)$$

$$+ 100\left(1 - \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}}\right) = 100 - 400\frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}} = 0$$

$$\Rightarrow \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}} = \frac{1}{4}$$

$$\frac{\partial l(\beta_0,\beta_1)}{\partial \beta_0} = \sum_{i=1}^{n} y_i - \frac{e^{\beta_0+\beta_1 x_i}}{1+e^{\beta_0+\beta_1 x_i}}$$

$$= 100\left(0 - \frac{e^{\beta_0}}{1+e^{\beta_0}}\right) + 300\left(0 - \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}}\right) + 300\left(1 - \frac{e^{\beta_0}}{1+e^{\beta_0}}\right)$$

$$+ 100\left(1 - \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}}\right) = 400 - 400\frac{e^{\beta_0}}{1+e^{\beta_0}} - 400\frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}}$$

$$= 0$$

$$\Rightarrow \frac{e^{\beta_0}}{1+e^{\beta_0}} + \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}} = 1$$

$$\Rightarrow \frac{e^{\beta_0}}{1+e^{\beta_0}} = 1 - \frac{1}{4} = \frac{3}{4} \;\Rightarrow\; \hat{\beta}_0 = \log\left(\frac{\frac{3}{4}}{1-\frac{3}{4}}\right) = \log 3$$

$$\hat{\beta}_0 + \hat{\beta}_1 = \log\left(\frac{\frac{1}{4}}{1-\frac{1}{4}}\right) = \log\frac{1}{3} \;\Rightarrow\; \hat{\beta}_1 = \log\frac{1}{3} - \log 3 = -\log 9$$

OR based on logistic regression model:

$$\widehat{OR}_{MLE} = \frac{\frac{\Pr(y=1|x=0)}{\Pr(y=0|x=0)}}{\frac{\Pr(y=1|x=1)}{\Pr(y=0|x=1)}} = e^{-\hat{\beta}_1} = e^{\log 9} = 9$$

Consistent with the odds ratio obtained in part (b).

(d) (3) Based on this dataset, can we estimate the probability of being alive for a female passenger? Explain your answer.

We cannot estimate this probability because the study is retrospective. The estimated coefficient $\hat{\beta}_0$ from the logistic regression model is biased for estimating absolute probabilities.